

VC BIRD INTL, Antigua and Barbuda

WMO: 788620

Lat: 17.137N

Lon: 61.793W

Elev: 19

StdP: 101.10

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	21.8	22.2	16.1	11.5	24.7	17.1	12.2	24.9	11.4	25.5	10.4	25.8	1.9	60	0.507

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.6	31.7	26.2	31.2	26.0	30.9	25.9	27.1	30.3	26.7	30.0	26.5	29.8	6.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.6	29.3	25.8	21.2	29.1	25.2	20.4	28.8	85.6	30.3	83.5	30.1	82.4	29.9	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.0	9.2	8.6	20.0	32.6	0.8	0.6	19.4	33.1	18.9	33.5	18.5	33.8	17.9	34.3
			17.4	27.9	0.9	1.0	16.7	28.6	16.2	29.1	15.6	29.7	15.0	30.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.1	25.6	25.5	25.9	26.4	27.2	28.0	28.3	28.5	28.1	27.8	27.1	26.2	(6)
	DBStd	1.37	0.79	0.76	0.88	0.85	0.97	0.84	0.82	0.80	0.92	0.96	0.96	0.90	(7)
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
	CDD10.0	6223	483	434	491	493	534	540	567	572	544	553	512	501	(10)
	CDD18.3	3182	224	201	233	243	276	290	308	314	294	294	262	242	(11)
	CDH23.3	31462	1596	1373	1752	2122	2784	3243	3581	3725	3413	3253	2597	2024	(12)
	CDH26.7	8127	152	146	247	391	666	952	1173	1323	1175	1013	609	280	(13)
(14) Wind	WSAvg	5.3	5.5	5.7	5.4	5.5	5.4	5.8	6.2	5.5	4.4	4.2	4.4	5.2	(14)
(15) Precipitation	PrecAvg	1176	66	41	43	72	87	96	116	141	146	148	134	86	(15)
	PrecMax	1823	134	84	103	252	246	203	246	258	323	270	320	232	(16)
	PrecMin	768	27	5	10	15	4	29	25	74	42	50	35	35	(17)
	PrecStd	246	24	16	21	48	64	45	50	48	64	66	84	45	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.2	29.5	30.0	30.2	31.4	31.2	32.0	32.1	32.1	32.0	31.1	30.0	(19)
		MCWB	23.7	23.3	23.8	24.2	25.3	25.8	26.6	26.8	26.0	26.2	25.7	25.4	(20)
	2%	DB	28.2	28.6	29.1	29.8	30.6	30.9	31.1	31.2	31.2	31.1	30.2	29.1	(21)
		MCWB	23.2	22.8	23.4	24.1	25.1	25.7	26.1	26.3	26.1	26.0	25.4	24.3	(22)
	5%	DB	28.0	28.1	28.5	29.1	30.0	30.2	30.7	31.0	31.0	30.7	29.9	28.8	(23)
		MCWB	23.1	22.6	23.1	23.8	24.9	25.3	25.8	26.1	26.1	25.8	25.2	24.0	(24)
	10%	DB	27.7	27.7	28.0	28.8	29.2	30.0	30.1	30.5	30.5	30.1	29.2	28.1	(25)
		MCWB	22.9	22.5	22.8	23.7	24.5	25.2	25.5	25.9	25.9	25.6	24.8	23.6	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.2	24.5	25.1	25.7	26.5	27.0	27.7	28.0	27.2	27.1	26.6	26.3	(27)
		MCDB	27.7	27.9	28.1	28.7	29.9	30.2	30.6	30.6	30.7	30.6	29.6	28.6	(28)
	2%	WB	24.3	23.7	24.3	25.0	25.9	26.5	26.8	27.1	26.7	26.6	26.1	25.5	(29)
		MCDB	27.3	27.2	27.7	28.4	29.2	29.7	29.8	30.2	30.2	30.1	29.2	28.0	(30)
	5%	WB	23.7	23.4	23.8	24.6	25.5	26.0	26.4	26.6	26.5	26.2	25.7	24.8	(31)
		MCDB	26.9	26.8	27.3	28.0	28.9	29.4	29.6	29.9	30.0	29.6	28.8	27.5	(32)
	10%	WB	23.4	23.0	23.4	24.1	25.1	25.6	26.0	26.3	26.2	25.9	25.3	24.3	(33)
		MCDB	26.6	26.4	27.0	27.6	28.5	29.0	29.3	29.6	29.6	29.2	28.5	27.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	4.5	4.8	4.9	4.9	4.6	4.2	4.3	4.6	5.0	5.0	4.8	4.5	(35)
		MCDBR	4.7	5.2	5.2	5.3	5.1	4.4	4.5	4.8	5.4	5.5	5.2	4.8	(36)
	5% WB	MCWBR	2.6	2.5	2.5	2.4	2.3	2.2	2.3	2.4	2.5	2.7	2.7	2.6	(37)
		MCWBR	4.5	5.0	5.1	5.1	4.8	4.3	4.5	4.8	5.1	5.1	5.0	4.6	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.363	0.370	0.371	0.403	0.442	0.520	0.512	0.476	0.432	0.389	0.372	0.369	(40)
		taud	2.603	2.575	2.571	2.503	2.407	2.134	2.157	2.300	2.464	2.602	2.639	2.616	(41)
	Ebn at Noon	Ebn at Noon	927	936	941	904	857	785	793	829	870	905	912	911	(42)
		Edh at Noon	93	100	103	111	120	156	153	134	113	96	89	90	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	4.98	5.69	6.37	6.60	6.39	6.41	6.43	6.41	5.96	5.35	4.83	4.63	(44)
	RadStd	RadStd	0.17	0.15	0.18	0.26	0.36	0.24	0.31	0.21	0.28	0.32	0.30	0.20	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	-0.28	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page